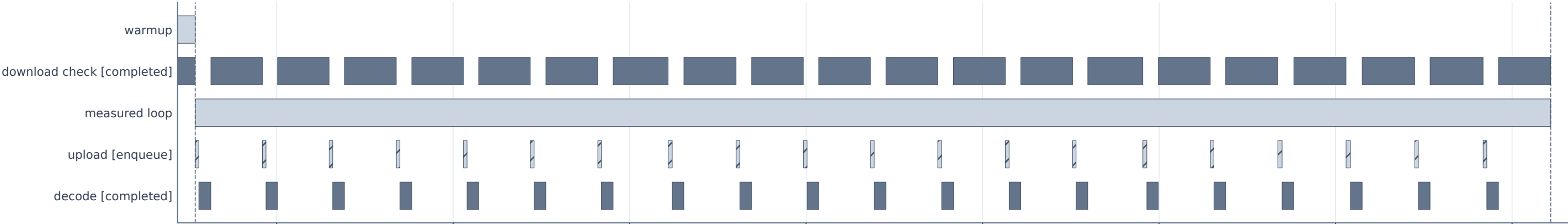


Warmed-loop CUDA timeline

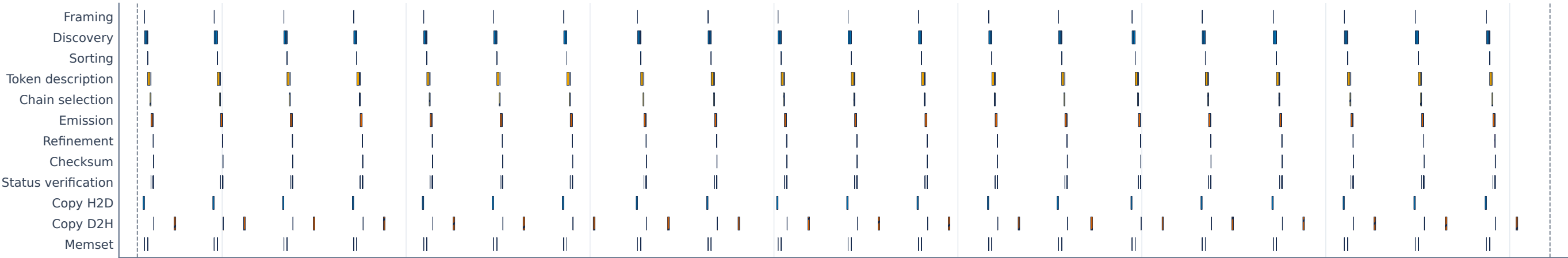
NVIDIA GeForce RTX 4090 · 64 MiB float32 · 20 validated measured iterations · source 6e49963961b2

Instrumented diagnostics · 2 warmups · synchronized panels share elapsed time; resource samples have lower temporal resolution

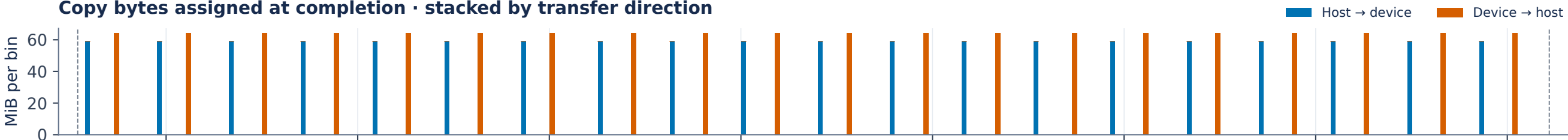
Host annotations · nested phases and repeated calls



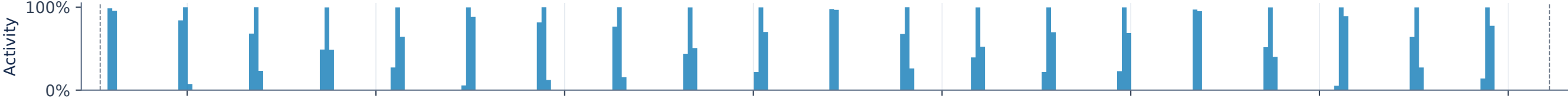
Exact recorded CUDA intervals · stage colors match the Nsight figures



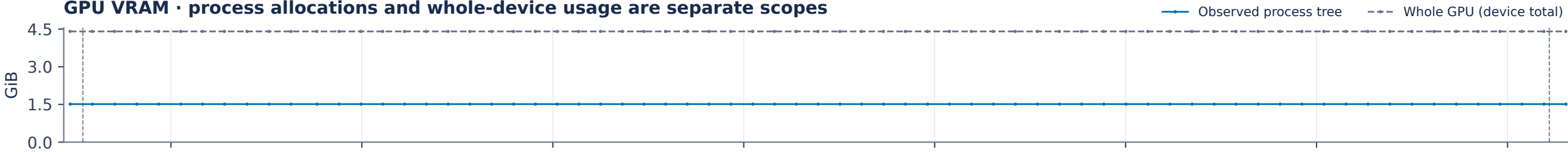
Copy bytes assigned at completion · stacked by transfer direction



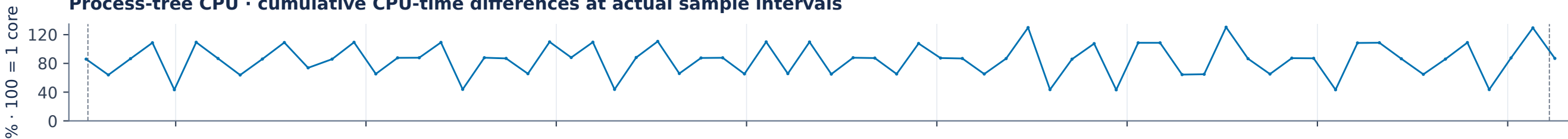
Recorded kernel activity fraction · interval union / bin duration; not occupancy



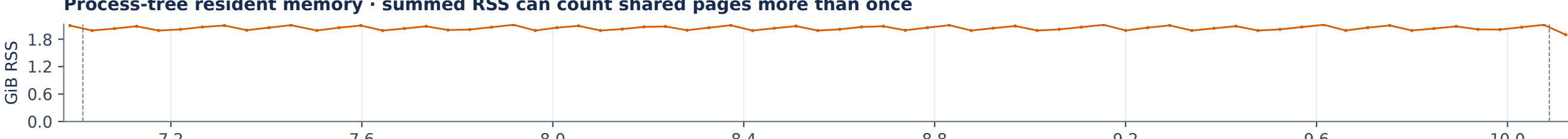
GPU VRAM · process allocations and whole-device usage are separate scopes



Process-tree CPU · cumulative CPU-time differences at actual sample intervals



Process-tree resident memory · summed RSS can count shared pages more than once



Elapsed time since collector clock origin (s)

Dashed lines bound the measured loop; whole-process shading locates it. CUDA bins: 10 ms; view-boundary bins use their actual clipped width.

Collector cadence: median 45.812 ms (44.870–50.833 ms). Metric points use query midpoints; CPU uses averaging-interval midpoints.

Host/CUDA alignment: $\pm 42.355 \mu\text{s}$ (half the origin-mark bracket). Host annotations do not establish GPU ownership or exclusive costs.

Transfers are completed bytes per bin, not instantaneous bandwidth. Coarse resource samples cannot resolve individual short phases; nulls remain gaps.

Nsight warns potentially missing NVTX events; all expected host phase ranges were validated. Other warnings remain in the report and manifest.